UNIVERSITY OF DHAKA

Department of Applied Mathematics

First Year B.S. (Honours) Program, Session: 2024 – 2025

Course Code: AMTH 150, Course Title: MATH LAB I (MATHEMATICA)

Assignment – 2 (Part A): Differential Calculus and its Real-life Applications

Instruction: Write programming code using MATHEMATICA to get the outputs and visualize the obtained results of the following problems.

Name: **Roll:** **Group:**

1. (a) Sketch the graphs of the following functions with their asymptotes using different colors:

$$(i)\, y=\frac{x^2+2x}{x^2-1}, (ii)\, y=\frac{x^2-1}{x^4+1}; -7\le x\le 7, (iii)\, y=\tan x, (iv)\, y=\csc x; -2\pi\le x\le 2\pi, (v)\, f(x)=e^{x+1}-2,$$

$$(vi)\, y=\sec^{-1}x, -1\le x\le 1, (vii)\, f(x)=1-2^x, (viii)\, y=\log(2-x^2), (ix)\, f(x)=1-|2-3x|$$

Also, find the domain and range in each case, and confirm that your obtained results are consistent with graphs.

(b) Find the formula for the inverse of $(i)\, f(x)=-\frac{1}{\sqrt{2x-3}}$ and $(ii)\, f(x)=2-\sqrt{1-3x}$, and compute the domain and range of f and f^{-1} in each case. Also, draw the graphs of $f(x)$ and $f^{-1}(x)$, and show that these graphs are reflections of one another about the line $y=x$.

(c) Justify whether the identity: $f\circ g=g\circ f$ is valid or not, where $f(x)=x^2+3$ and $g(x)=\sqrt{x}$. Explain your answer.

(d) The projected world population (in billions of people) t years after 2017 is described by the function:

$$f(t)=0.0729e^{0.45t}.$$

(i) Based on this model what will the world population be in 2031?

(ii) If this trend continues, approximately when will the world population reach 19 billion?

2. (a) Find the limit of $f(x)=\begin{cases}1/(x+2), & x<-2\\ x^2-5, & -2<x\le 3\\ \sqrt{x+13}, & x>3\end{cases}$ as $(i)\, x\to -2, (ii)\, x\to 0, (iii)\, x\to 3$, and verify that the limit calculations in **parts (i), (ii), and (iii)** are consistent with the graph of $f(x)$.

(b) Suppose that the speed v (in ft/s) of a skydiver t seconds after leaping from a plane is given by the equation $v=190\left(1-e^{-0.168t}\right)$.

(i) Graph v versus t.

(ii) By evaluating an appropriate limit, show that the graph of v versus t has a horizontal asymptote $v=c$ for an appropriate constant c.

(iii) What is the physical significance of the constant c in **part (ii)**?

3. (a) Determine the points, if any, at which the following functions are discontinuous, and sketch their graphs, showing the points of discontinuity.

$$(i)\, f(x)=\frac{x^3+8}{x^3+3x^2-4x-12}, (ii)\, f(x)=\begin{cases}\frac{\sin x}{x}, x\ne 0\\ \frac{1}{2}, \quad x=0\end{cases}, (iii)\, f(x)=\begin{cases}0, x<-1\\ \sqrt{1-x^2}, -1<x<1\\ x, x\ge 1\end{cases}.$$

(b) For what values of the constants k and m, if possible, that will make the function $f(x)=\begin{cases} x^2+5, & x>2 \\ m(x+1)+k, & -1<x\le 2 \\ 2x^3+x+7, & x\le -1 \end{cases}$ continuous everywhere, and verify the obtained results with the graph of $f(x)$.

(c) Find the values of α and β, if possible, so that the function $f(x)=\begin{cases} 3x^2, & x\le 1 \\ \alpha x+\beta, & x>1 \end{cases}$ will be differentiable.

4. (a) Find the derivative with respect to x of $f(x)=\sqrt{x}$ and use it to find the equation of tangent line to $y=\sqrt{x}$ at $x=9$. Also, find the limit of $f'(x)$ as $x\to 0^+$ and as $x\to+\infty$, and explain what those limits say about the graph of f.

(b) Find an equation for the tangent line to the curve $x^3-y^3=3xy$ at the point $(-3/2,3/2)$. At what point(s) in the second quadrant is the tangent line to the curve vertical? Confirm that your obtained results are consistent with graphs and asymptote.

(c) Sketch and identify curve defined by $x^{2/3}+y^{2/3}=4$. Also, use *implicit differentiation* to find the slope of the tangent to the curve at the point $\left(-1,3\sqrt{3}\right)$ and visualize in the graph.

(d) Apply the following *tabular data* to compute $F'(-1)$, where $F(x)=[(g\circ f)(x)]^3$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	3	2	-3
2	-1	4	1	-5

5. (a) Apply the $L'H\hat{o}pital's$ Rule to find the limits:

$$(i)\lim_{\theta\to 1}\left(2-\theta\right)^{\tan[(\pi/2)\theta]} \quad \text{and} \quad (ii)\lim_{y\to 0}\left(\frac{1}{y}-\frac{1}{e^y-1}\right).$$

(b) A 10 ft plank is leaning against a wall. If at a certain instant the bottom of the plank is 2 ft from the wall and is being pushed toward the wall at the rate of 6 in/s, how fast is the acute angle that the plank makes with the ground increasing? Comment on your obtained result.

(c) Wheat is poured through a chute at the rate of 10 ft^3/min and falls in a conical pile whose bottom radius is always half the altitude. How fast will the circumference of the base be increasing when the pile is 8 ft high? Comment on your obtained result.

6. (a) Consider the functions: $(i)\, f(x)=\frac{1}{2}x^4-\frac{4}{3}x^3-x^2+4x+1, (ii)\, f(x)=x+2\sin x$. Determine the intervals on which these functions are increasing, decreasing, concave up, and concave down. Confirm that your obtained results are consistent with graphs of $f(x), f'(x),$ and $f''(x)$ in the same coordinate axes. Also, draw $f(x)$ with all its critical points, inflection points, relative extremum points in the same graph.

(b) In the interval $[-1,2]$, determine the absolute maximum and minimum values of the function $f(t)=\begin{cases} 6t-5, t>1 \\ 4t^2+3t-7, t\le 1 \end{cases}$, and round them to the nearest hundredth. Graphically illustrate the function's absolute extremum values with heavy red dots.

(c) An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume? Confirm that the obtained results are consistent with the graph.

7. (a) Construct the Maclaurin polynomials of degree 1,3,5,7 for $(i)\cos ec(x)-1$ and $(ii)\cos x$; and compute their value at $x=\dfrac{\pi}{4}$. Determine the error in the approximation and express it in a tabular form. Plot the polynomials together with the given functions on the same diagram with different colors, and observe the relation between the given function with the resultant polynomials.

(b) Construct the first four Taylor polynomials for $(i)\ln(x)$ about $x=2$, $(ii)e^{-2x}$ about $x=0$, and $(iii)\sin x$ about $x=\dfrac{\pi}{2}$; and use a graphing utility to graph the given functions and the Taylor polynomials on the same coordinate system.

8. (a) Consider the function: $f(x)=\dfrac{1}{4}x^3+1$.

(i) Find the equation of the secant line through the points $(0, f(0))$ and $(2, f(2))$.

(ii) Show that there is only one-point ξ in the interval $(0,2)$ that satisfies the Mean-Value Theorem.

(iii) Find the equation of the tangent line to the graph of f at the point $(c, f(c))$.

(iv) Show that the secant line in **part (i)** and the tangent line in part **(iii)** in the same coordinate system together with $f(x)$, and confirm visually that the two lines seem parallel.

(b) Justify whether or not the function $f(x)=\log(-x^2+2x+4)$ satisfies the hypotheses of Rolle's theorem on the interval $[-1,3]$. If so, find all values of ξ that satisfy the conclusion of the theorem. Visualize the obtained results.

The End